

"Ridgeline HD 6.7 TD: Diesel particulate filter efficiency below threshold"

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Area: bulletins

1 Condition

Affected Ridgeline HD 6.7 TD vehicles may set DTC P2002 (Diesel Particulate Filter Efficiency Below Threshold, Bank 1) with the MIL illuminated and, in some cases, a reduced-power message. The DPF-5180 diesel particulate filter shows abnormal soot loading and an elevated differential pressure reading that does not recover after normal driving. Customers commonly report the condition after extended low-speed or short-trip operation that prevents passive regeneration. Confirm P2002 is current and review the differential-pressure data per the Ridgeline HD Emissions Service Manual :: Diagnosis section.

2 Cause

Incomplete regeneration cycles allow soot to accumulate in the DPF-5180 beyond the calibrated efficiency window, and the differential-pressure model interprets the restriction as a degraded filter, setting P2002. In many cases the DPF-5180 is mechanically sound and can be restored by a forced regeneration rather than replacement. A cracked or melted substrate, however, requires filter replacement and cannot be regenerated. The fault-setting logic for P2002 is detailed in the Ridgeline HD Emissions Service Manual :: Emissions Diagnostic Trouble Code Reference .

3 Diagnostic Verification

Using the scan tool, record the DPF-5180 differential pressure and estimated soot load with the engine at operating temperature, then compare against the published thresholds before condemning the filter. If P2002 is accompanied by a measured backpressure above the substrate-damage limit, replace the DPF-5180; otherwise proceed to forced regeneration. Rule out exhaust leaks and a faulty differential-pressure sensor, both of which can mimic P2002. The full verification flow is documented in the Ridgeline HD Emissions Service Manual :: Emissions System Overview and the Ridgeline HD Emissions Service Manual :: Diesel Particulate Filter Sub-System .

4 Repair Procedure

Perform a scan-tool-commanded forced regeneration of the DPF-5180 following PROC-DPF-REGEN, observing the exhaust-temperature and soot-load targets throughout the cycle. After regeneration completes, clear the codes and confirm the soot load and differential pressure have returned to normal. If the soot load fails to drop or P2002 returns, replace the DPF-5180 as directed. The complete forced-regeneration sequence and safety precautions are in the Ridgeline HD Emissions Service Manual :: Diesel Particulate Filter Service and Forced Regeneration .